SCRUTINIZING COLUMN-ORIENTED DATABASES

OBJECTIVES

This is time to get familiar with the internals of column-oriented databases.

A column-oriented database uses vertical fragmentation to store data. Thus, opposite to row-oriented databases (i.e., traditional relational databases) that store data by rows, they store it by columns. As consequence, in front of random analytical queries (e.g., those of a Data Warehouse) it tends to yield a much better **effective read ratio**.

In this lab we will scrutinize the differences between row-oriented and column-oriented databases. Although row-oriented database vendors claim they can achieve similar performance with the correct tuning, this turns not to be true.

In this lab you are asked to simulate, in paper, how column-oriented databases would store data and execute queries. This lab is done in paper since column-oriented databases can be fully relational (or follow other data models as we will see later in the course) and the interface is SQL. All the materials we are discussing in the lectures remain in the internals and therefore not seen by the end-users when *using* the database.

The list of objectives for this session is as follows:

- Given a real database and a read-only workload:
    - Store data in a columnar fashion following the run-length encoding,
    - Execute queries following the vectorized query processing principles.
- In parallel, you should understand how different the storage and execution of those queries would be in a traditional row-oriented database.

REQUIRED KNOWLEDGE

This lab assumes attendance or understanding of the materials presented in the column-oriented theory session.
Your lab mates for this session will be that of the team creation 3 event.

TOOLS

This lab is done in paper.

DELIVERABLES

The main task for this lab is to solve the Learn-SQL *3rd Lab: Scrutinizing Column-Oriented Databases*.

Solve the exercise and upload a document with your solution to Learn-SQL to the *3rd Lab Document Submission* (assignment).

OPTIONAL ACTIVITIES

Repeat these exercises in a column-oriented database:

- Go here and download MonetDB, an open-source column-oriented database.
- Implement the schema proposed in the exercise and insert a few thousands of (random) rows and launch the queries.
- Ask MonetDB for **the execution plan**. This is a real column-oriented database! You will see the differences with the relational execution discussed in the lab.
- If really interested, implement the lab also in PostgreSQL and ask for the execution plan and see the differences on the operations of both DBMS!